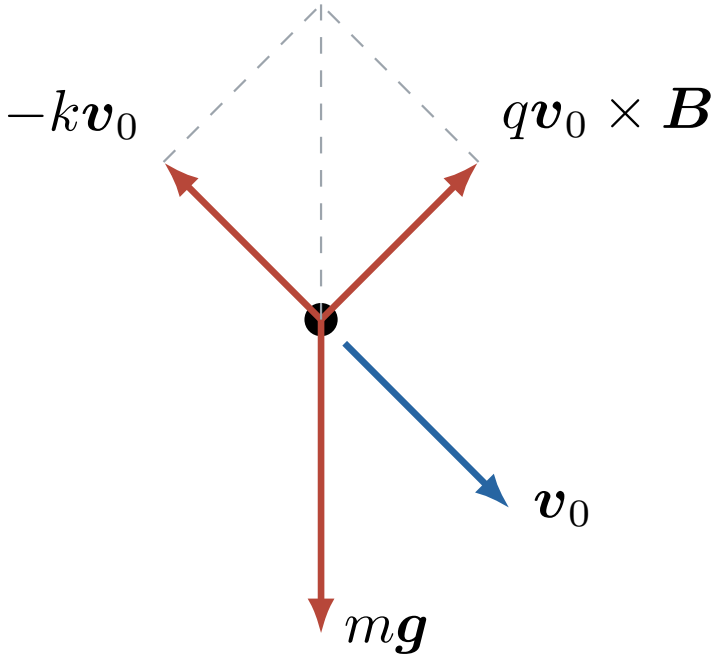
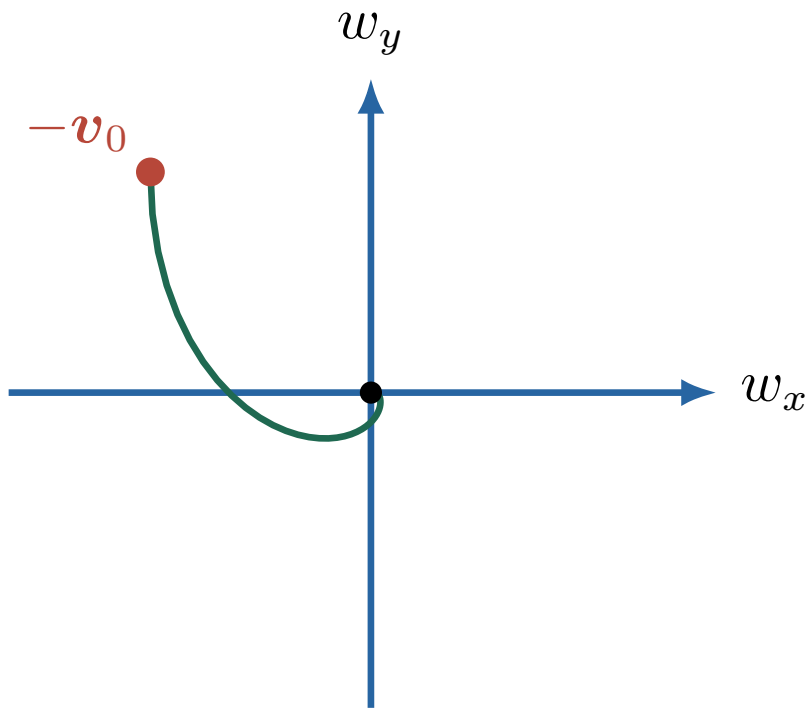


$qB = k > 0$



$|\boldsymbol{w}| = v_0 e^{-kt/m}$